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(54) **HETEROCYCLE SUBSTITUTED PYRIDINE DERIVATIVE ANTIFUNGAL AGENTS**

HETEROZYKLUS-SUBSTITUIERTE PYRIDINDERIVATANTIMYKOTIKA

AGENTS ANTIFONGIQUES DÉRIVÉS DE PYRIDINE SUBSTITUÉS PAR UN HÉTÉROCYCLE

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(56) References cited:

EP-A1- 1 944 303 EP-A1- 2 345 328
EP-A1- 2 628 738 WO-A1-2020/005860
WO-A2-2011/051198 US-A1- 2006 247 237
US-A1- 2010 004 235 US-A1- 2010 168 173

- **SHAW KAREN JOY ET AL: "In Vitro and In Vivo Evaluation of APX001A/APX001 and Other Gwt1 Inhibitors against Cryptococcus", ANTIMICROBIAL AGENTS AND CHEMOTHERAPY, vol. 62, no. 8, 1 August 2018 (2018-08-01), XP055871494, US ISSN: 0066-4804, DOI: 10.1128/AAC.00523-18 Retrieved from the Internet: URL:https://journals.asm.org/doi/pdf/10.1128/AAC.00523-18>**

Remarks:

The file contains technical information submitted after the application was filed and not included in this specification

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EP 3 720 438 B1

Description

BACKGROUND

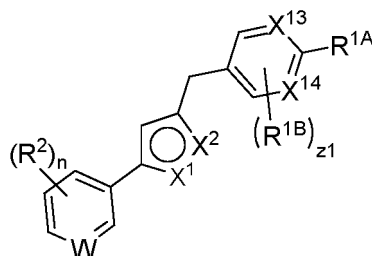
[0001] A need exists in the art for an effective treatment of fungal diseases.

[0002] The following patent publications describe pyridine derivatives which are said to be useful as antifungal agents: international patent publication WO-2020/005860; European patent publication EP 2 628 738 A1; European patent publication EP 1 944 303 A1; international patent publication WO-2011/051198; European patent publication EP 2 345 328 A1.

BRIEF SUMMARY OF THE INVENTION

[0003] Provided herein is the compound 3-(3-(4-((6-fluoropyridin-2-yl)oxy)benzyl)isoxazol-5-yl)pyridin-2-amine, or a pharmaceutically acceptable salt, solvate, hydrate, tautomer, or an isotopic variant thereof, and a pharmaceutical composition comprising said compound. The subject compound and composition are useful for treating fungal diseases. Furthermore, the subject compound and composition are useful for the treatment of cryptococcosis.

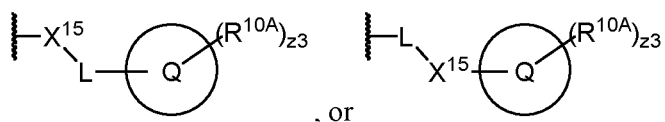
[0004] The compound 3-(3-(4-((6-fluoropyridin-2-yl)oxy)benzyl)isoxazol-5-yl)pyridin-2-amine is Example 111 and falls within the scope of Formula (III):



Formula (III);

wherein:

R^{1A} is -OH, substituted or unsubstituted C₁₋₆ alkyl, -(CH₂)₂S(CH₂)₂OC(O)H, -X¹⁵-L-C≡N, -L-X¹⁵-(CH₂)₂₂C≡N, -X¹⁵-L-CH=CR²⁹R³⁰,



one of X¹ and X² is N while the other is O;

X¹³ and X¹⁴ are independently N or C(R^{1B});

X¹⁵ is a bond, -NH-, -O-, -S-, or -SO₂-;

L is a bond or substituted or unsubstituted C₁₋₆ alkylene;

W is N or N⁺-OPO₃H⁻;

Ring Q is cycloalkyl, heterocycloalkyl, aryl, or heteroaryl;

each R² is independently hydrogen, -NH₂, or halogen;

each R^{1B} and R^{10A} is independently hydrogen, halogen, -CF₃, -CN, -CH₂-OH, -OR^a, -SR^a, -S(=O)R^b, -NO₂, -NR^cR^d, -S(=O)₂R^d, -NR^aS(=O)₂R^d, -S(=O)₂NR^cR^d, -C(=O)R^b, -OC(=O)R^b, -CO₂R^a, -OCO₂R^a, -C(=O)NR^cR^d, -OC(=O)NR^cR^d, -NR^aC(=O)NR^cR^d, -NR^aC(=O)R^b, -NR^aC(=O)OR^a, substituted or unsubstituted C₁₋₆ alkyl, substituted or unsubstituted, substituted or unsubstituted C₂₋₆ alkenyl, substituted or unsubstituted C₂₋₆ alkynyl, substituted or unsubstituted C₁₋₆ heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl;

R²⁹ and R³⁰ are independently hydrogen, halogen, substituted or unsubstituted C_i-C_e alkyl, substituted or unsubstituted aryl, substituted or unsubstituted heteroaryl, substituted or unsubstituted cycloalkyl, or substituted or unsubstituted heterocycloalkyl;

R^a is hydrogen, substituted or unsubstituted C₁₋₆ alkyl, substituted or unsubstituted C₂₋₆ alkenyl, substituted or unsubstituted C₂₋₆ alkynyl, substituted or unsubstituted C₁₋₆ heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted het-

heteroaryl;

R^b is substituted or unsubstituted C₁-C₆ alkyl, substituted or unsubstituted C₂-C₆ alkenyl, substituted or unsubstituted C₂-C₆ alkynyl, substituted or unsubstituted C₁-C₆ heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl;

R^c and R^d is independently hydrogen, substituted or unsubstituted C₁-C₆ alkyl, substituted or unsubstituted C₂-C₆ alkenyl, substituted or unsubstituted C₂-C₆ alkynyl, substituted or unsubstituted C₁-C₆ heteroalkyl, substituted or unsubstituted cycloalkyl, substituted or unsubstituted heterocycloalkyl, substituted or unsubstituted aryl, or substituted or unsubstituted heteroaryl;

n is 0-3;

z₁ is 0-2;

z₂ is 0-3; and

z₃ is 1-3.

Only Example 111 is a compound of the invention. All other "Examples" are merely reference examples.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005]

FIG. 1 shows the efficacy of certain compounds of the present disclosure in a murine model of cryptococcal meningitis when dosed in the presence of the pan-CYP inhibitor 1-aminobenzotriazole (ABT).

FIG. 2 shows a dose response study with (2-amino-3-(3-(4-((6-fluoropyridin-2-yl)oxy)benzyl)isoxazol-5-yl)pyridin-1-ium-1-yl)methyl hydrogen phosphate (**Example 172**) and (2-amino-3-(3-(4-(benzyloxy)benzyl)isoxazol-5-yl)pyridin-1-ium-1-yl)methyl hydrogen phosphate (**Example 173**).

FIG. 3 shows the efficacy of (2-amino-3-(3-(4-((6-fluoropyridin-2-yl)oxy)benzyl)isoxazol-5-yl)pyridin-1-ium-1-yl)methyl hydrogen phosphate (**Example 172**) and amphotericin B (AMB) in a delayed model of cryptococcal meningitis.

DETAILED DESCRIPTION OF THE INVENTION

[0006] The incidence of fungal infections has increased over the last few decades. Such infections have risen in the last few decades in part due to an increase in individuals that are immunocompromised. Immunocompromised individuals, include, for example, elderly individuals, individuals with HIV/AIDS, or individuals undergoing chemotherapy treatment or immunosuppressive therapy after a transplant.

[0007] Current antifungal therapies exploit differences between mammalian cells and fungal cells to kill the fungi. However, because fungi and mammals are both eukaryotes, many antifungal therapies cause side effects in the host mammal. Additionally, many fungal organisms have developed resistance to front line antifungal treatments. Thus, there exists a need for new compositions and methods for treating fungal diseases.

Definitions

[0008] When ranges are used herein for physical properties, such as molecular weight, or chemical properties, such as chemical formulae, all combinations and subcombinations of ranges and specific embodiments therein are intended to be included. The term "about" when referring to a number or a numerical range means that the number or numerical range referred to is an approximation within experimental variability (or within statistical experimental error), and thus the number or numerical range, in some instances, will vary between 1% and 15% of the stated number or numerical range. The term "comprising" (and related terms such as "comprise" or "comprises" or "having" or "including") is not intended to exclude that in other certain embodiments, for example, an embodiment of any composition of matter, composition, method, or process, or the like, described herein, "consist of" or "consist essentially of" the described features.

[0009] As used in the specification and appended claims, unless specified to the contrary, the following terms have the meaning indicated below.

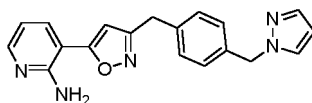
[0010] "Alkyl" refers to an substituted or unsubstituted straight-chain, or substituted or unsubstituted branched-chain saturated hydrocarbon monoradical having from one to about ten carbon atoms, or from one to six carbon atoms, wherein a sp³-hybridized carbon of the alkyl residue is attached to the rest of the molecule by a single bond. Examples include, but are not limited to, methyl, ethyl, n-propyl, isopropyl, 2-methyl-1-propyl, 2-methyl-2-propyl, 2-methyl-1-butyl, 3-methyl-1-butyl, 2-methyl-3-butyl, 2,2-dimethyl-1-propyl, 2-methyl-1-pentyl, 3-methyl-1-pentyl, 4-methyl-1-pentyl, 2-methyl-2-pentyl, 3-methyl-2-pentyl, 4-methyl-2-pentyl, 2,2-dimethyl-1-butyl, 3,3-dimethyl-1-butyl, 2-ethyl-1-butyl, n-butyl, isobutyl, sec-butyl, t-butyl, n-pentyl, isopentyl, neopentyl, tert-amyl and hexyl, and longer alkyl groups, such as heptyl, octyl, and

[0121] 2-fluoro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine (1.00g, 4.44mmol) and di-tert-butyl [3-(3-(chloromethyl)-1,2-oxazol-5-yl)pyridin-2-yl]imidodicarbonate (Intermediate A, 2.00g, 4.88mmol) were mixed with DME (15mL) in a sealable tube. A 2M solution of sodium carbonate in water (5.55mL, 11.09mmol) and palladium tetrakis triphenylphosphine (0.36g, 0.31mmol) were added and the sealable tube was flushed with argon and sealed. The mixture was stirred for 2h at 95°C. The cooled reaction mixture was poured into ethyl acetate (500mL), dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (SiO₂, hexane/ethyl acetate) to give the di-Boc protected coupling intermediate, to which was added formic acid (8mL). The resulting mixture was stirred for 13h at 21-25°C to complete the di-Boc de-protection. During this step the fluoropyridine also completely hydrolyzed. Toluene (100mL) and acetonitrile (50mL) were added and all volatiles were removed under reduced pressure. This addition/evaporation procedure was repeated three times to complete the removal of formic acid and to obtain a solid residue of 5-((5-(2-aminopyridin-3-yl)isoxazol-3-yl)methyl)pyrimidin-2-ol (570mg, 2.12mmol). The residue was suspended in acetonitrile (2mL). N-benzyl-N,N-diethylethanaminium chloride (237mg, 1.04mmol) and phosphoryl trichloride (0.58mL, 6.24mmol) were added, followed by the addition of N-ethyl-N-isopropylpropan-2-amine (0.71mL, 4.16mmol). The mixture was heated in a sealed tube at 90°C for 20h and then slowly added to a stirred solution of NaHCO₃ in water. Some ice was added to maintain a temperature around 30°C. After quenching was complete, the mixture was extracted with EtOAc. The combined organic layers were dried and concentrated under reduced pressure to obtain the crude product (3-(3-((2-chloropyrimidin-5-yl)methyl)isoxazol-5-yl)pyridin-2-amine (180mg, 0.63mmol, 13% overall yield)) as a solid, which was directly used in the following experiments without further purification. 500 MHz ¹H NMR (DMSO-d₆) δ 8.80 (s, 2H), 8.10 (dd, J = 4.8, 1.9 Hz, 1H), 7.86 (dd, J = 7.7, 1.9 Hz, 1H), 6.89 (s, 1H), 6.71 (dd, J = 7.7, 4.8 Hz, 1H), 6.28 (s, 2H), 4.15 (s, 2H). MS: 287.9 [M+H]⁺.

EXAMPLES

Example 1 (for reference only): Synthesis of 3-(3-(4-((1H-pyrazol-1-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

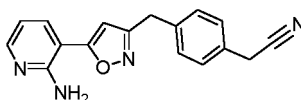
[0122]



[0123] A solution of 1H-pyrazole (91mg, 1.33mmol) in NMP (0.5mL) was added to a suspension of sodium hydride (60% w/mineral oil, 40mg, 1.00mmol) in NMP (1mL). After stirring for 20min at 21-25°C, a solution of 3-(3-(4-(chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 100mg, 0.33mmol) in NMP (1mL) was added and the mixture was stirred for 4min at 60°C. The cooled reaction mixture was directly purified by column chromatography (SiO₂, hexane/ethyl acetate). Fractions containing the product were concentrated under reduced pressure and further purified by HPLC to yield 3-(3-(4-((1H-pyrazol-1-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (75mg, 0.23mmol, 68%) as a white solid. 400 MHz ¹H NMR (CDCl₃) δ 8.12 (dd, J = 4.9, 1.8 Hz, 1H), 7.68 (dd, J = 7.7, 1.8 Hz, 1H), 7.54 (d, J = 1.8 Hz, 1H), 7.39 (dd, J = 2.3, 0.7 Hz, 1H), 7.29 - 7.21 (m, 2H), 7.21 - 7.13 (m, 2H), 6.69 (dd, J = 7.7, 4.9 Hz, 1H), 6.31 - 6.20 (m, 2H), 5.42 (s, 2H), 5.30 (s, 2H), 4.03 (s, 2H). MS: 332.2 [M+H]⁺.

Example 2 (for reference only): Synthesis of 2-(4-((5-(2-aminopyridin-3-yl)isoxazol-3-yl)methyl)phenyl)acetonitrile

[0124]

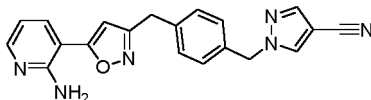


[0125] Potassium cyanide (209mg, 3.20mmol) was added to PEG400 (1.5mL) and carefully ground to a fine suspension using a glass rod. 3-(3-(4-(chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 120mg, 0.40mmol) was added and the mixture was stirred for 30min at 80°C. To the cooled reaction mixture was added to water (50 mL), and then extracted with ethyl acetate (3×30 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (SiO₂, hexane/ethyl acetate) to yield 2-(4-((5-(2-aminopyridin-3-yl)isoxazol-3-yl)methyl)phenyl)acetonitrile (102mg, 0.35mmol, 88%) as a white solid. 400 MHz ¹H NMR (CDCl₃) δ 8.09 (dd, J = 4.9, 1.8 Hz, 1H), 7.65 (dd, J = 7.7, 1.8 Hz, 1H), 7.26

(m, 4H), 6.66 (dd, $J = 7.7, 4.9$ Hz, 1H), 6.20 (s, 1H), 5.38 (s, 2H), 4.01 (s, 2H), 3.69 (s, 2H). MS: 291.2 [M+H]⁺.

Example 3 (for reference only): Synthesis of 1-(4-((5-(2-aminopyridin-3-yl)isoxazol-3-yl)methyl)benzyl)-1H-pyrazole-4-carbonitrile

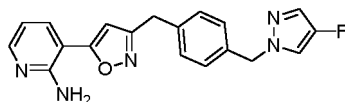
[0126]



[0127] 3-(3-(4-(chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 80mg, 0.27mmol) and 1H-pyrazole-4-carbonitrile (99mg, 1.07mmol) were dissolved in NMP (1ml). Potassium 2-methylpropane-2-olate (1M in THF, 0.80mL, 0.80mmol) was added and the mixture was stirred for 5min at 60°C. The cooled reaction mixture was directly purified by column chromatography (SiO₂, hexane/ethyl acetate). Fraction containing the product were concentrated under reduced pressure and further purified by HPLC to yield 1-(4-((5-(2-aminopyridin-3-yl)isoxazol-3-yl)methyl)benzyl)-1H-pyrazole-4-carbonitrile (66mg, 0.19mmol, 69%) as a white solid. 400 MHz ¹H NMR (CDCl₃) δ 8.13 (dd, $J = 5.0, 1.8$ Hz, 1H), 7.79 (dd, $J = 18.1, 0.7$ Hz, 2H), 7.69 (dd, $J = 7.7, 1.8$ Hz, 1H), 7.35 - 7.25 (m, 2H), 7.27 - 7.17 (m, 2H), 6.70 (dd, $J = 7.7, 4.8$ Hz, 1H), 6.25 (s, 1H), 5.43 (s, 2H), 5.30 (s, 2H), 4.05 (s, 2H). MS: 357.3 [M+H]⁺.

Example 4 (for reference only): Synthesis of 3-(3-(4-((4-fluoro-1H-pyrazol-1-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

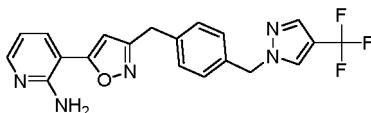
[0128]



[0129] 3-(3-(4-(chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 80mg, 0.27mmol) and 4-fluoro-1H-pyrazole (92mg, 1.07mmol) were dissolved in NMP (1ml). Potassium 2-methylpropane-2-olate (1M in THF, 0.80mL, 0.80mmol) was added and the mixture was stirred for 5min at 60°C. The cooled reaction mixture was directly purified by column chromatography (SiO₂, hexane/ethyl acetate). Fraction containing the product were concentrated under reduced pressure and further purified by HPLC to yield 3-(3-(4-((4-fluoro-1H-pyrazol-1-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (75mg, 0.22mmol, 80%) as a white solid. 400 MHz ¹H NMR (CDCl₃) δ 8.14 (s, 1H), 7.69 (dd, $J = 7.7, 1.7$ Hz, 1H), 7.35 (dd, $J = 4.3, 0.8$ Hz, 1H), 7.31 - 7.13 (m, 5H), 6.70 (dd, $J = 7.7, 4.7$ Hz, 1H), 6.24 (s, 1H), 5.43 (s, 2H), 5.18 (s, 2H), 4.04 (s, 2H). MS: 350.3 [M+H]⁺.

Example 5 (for reference only): Synthesis of 3-(3-(4-((4-(trifluoromethyl)-1H-pyrazol-1-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

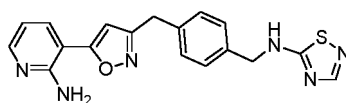
[0130]



[0131] 3-(3-(4-(chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 80mg, 0.27mmol) and 4-(trifluoromethyl)-1H-pyrazole (145mg, 1.07mmol) were dissolved in NMP (1ml). Potassium 2-methylpropane-2-olate (1M in THF, 0.80mL, 0.80mmol) was added and the mixture was stirred for 5min at 60°C. The cooled reaction mixture was directly purified by column chromatography (SiO₂, hexane/ethyl acetate). Fraction containing the product were concentrated under reduced pressure and further purified by HPLC to yield 3-(3-(4-((4-(trifluoromethyl)-1H-pyrazol-1-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (75mg, 0.19mmol, 70%) as a white solid. 400 MHz ¹H NMR (CDCl₃) δ 8.14 (d, $J = 3.7$ Hz, 1H), 7.73 (s, 1H), 7.69 (dd, $J = 7.7, 1.8$ Hz, 1H), 7.64 (s, 1H), 7.30 (d, $J = 8.2$ Hz, 2H), 7.22 (d, $J = 8.2$ Hz, 2H), 6.70 (dd, $J = 7.7, 4.9$ Hz, 1H), 6.25 (s, 1H), 5.43 (s, 2H), 5.29 (s, 2H), 4.05 (s, 2H). MS: 400.3 [M+H]⁺.

Example 6 (for reference only): Synthesis of N-(4-((5-(2-aminopyridin-3-yl)isoxazol-3-yl)methyl)benzyl)-1,2,4-thiadiazol-5-amine

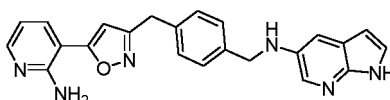
[0132]



[0133] 3-(3-(4-(Chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 0.10 g, 0.33 mmol) and 1,2,4-thiadiazol-5-amine (0.17 g, 1.7 mmol) were dissolved in tetrahydrofuran. Diisopropylethylamine (0.14 mL, 0.80 mmol) was added and the homogeneous amber-colored solution was stirred at room temperature. After 3 hours an additional 2 equivalents of 1,2,4-thiadiazol-5-amine (0.070 g, 0.66 mmol) and another 1.2 equivalents of diisopropylethylamine (0.070 mL, 0.40 mmol) were added. The mixture was heated at 60 °C for 72 hours. The mixture was cooled to room temperature and extracted three times with ethyl acetate. The organic phase was washed with brine, dried over sodium sulfate and evaporated under reduced pressure. The obtained residue was purified using Biotage reverse phase flash chromatography (12g C18 SNAP, 5-95% acetonitrile in water with 0.1% formic acid). The desired fraction was lyophilized to give the title compound as a white solid (2.0 mg, 0.0051 mmol, 2%). MS: 365.2 [M+H]⁺.

Example 7 (for reference only): Synthesis of N-(4-((5-(2-aminopyridin-3-yl)isoxazol-3-yl)methyl)benzyl)-1H-pyrrolo[2,3-b]pyridin-5-amine

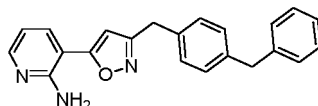
[0134]



[0135] 3-(3-(4-(Chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 0.10 g, 0.33 mmol) and 1H-pyrrolo[2,3-b]pyridin-5-amine (0.22 g, 1.7 mmol), were dissolved in tetrahydrofuran. Diisopropylethylamine (0.14 mL, 0.80 mmol) was added and the heterogeneous mixture was stirred at room temperature for 3 hours, and then heated at 60 °C for 72 hours. The mixture was cooled to room temperature and extracted three times with ethyl acetate. The organic phase was washed with brine, dried over sodium sulfate and evaporated under reduced pressure. The obtained residue was purified using Biotage reverse phase flash chromatography (12g C18 SNAP, 5-95% acetonitrile in water with 0.1% formic acid). The desired fraction was lyophilized to give the title compound as a white solid (20 mg, 0.051 mmol, 15%). MS: 397.3 [M+H]⁺.

Example 8 (for reference only): Synthesis of 3-(3-(4-benzylbenzyl)isoxazol-5-yl)pyridin-2-amine

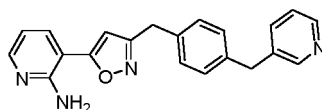
[0136]



[0137] Phenylboronic acid (61mg, 0.50mmol) and 3-(3-(4-(chloromethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (Intermediate B, 100mg, 0.33mmol) were mixed in DME (4mL) in a sealable tube. A 2M solution of sodium carbonate in water (0.45mL, 0.90mmol) and palladium tetrakis triphenylphosphine (27mg, 0.023mmol) were added and the sealable tube was flushed with argon and sealed. The mixture was stirred for 2h at 90°C. The cooled reaction mixture was poured into ethyl acetate and dried over Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by column chromatography (SiO₂, hexane/ethyl acetate) to yield 3-(3-(4-benzylbenzyl)isoxazol-5-yl)pyridin-2-amine (84mg, 0.25mmol, 74%) as a white solid. 400 MHz ¹H NMR (CDCl₃) δ 8.16 - 8.10 (m, 1H), 7.69 (dd, J = 7.7, 1.7 Hz, 1H), 7.33 - 7.12 (m, 9H), 6.69 (dd, J = 7.7, 4.8 Hz, 1H), 6.24 (s, 1H), 5.40 (s, 2H), 4.01 (s, 2H), 3.96 (s, 2H). MS: 342.2 [M+H]⁺.

Example 9 (for reference only): Synthesis of 3-(3-(4-(pyridin-3-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine

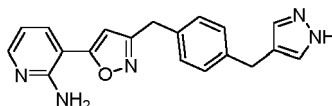
[0138]



[0139] The title compound was prepared according to the procedure described in Example 8 using pyridin-3-ylboronic acid (61.5mg, 0.50mmol) to yield 3-(3-(4-(pyridin-3-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (64mg, 0.19mmol, 56%) as a white solid. 400 MHz ^1H NMR (CDCl_3) δ 8.52 (s, 2H), 8.06 (d, J = 4.1 Hz, 1H), 7.73 (dd, J = 7.7, 1.7 Hz, 1H), 7.52 (d, J = 7.7 Hz, 1H), 7.30 - 7.24 (m, 1H), 7.22 (d, J = 8.1 Hz, 2H), 7.14 (d, J = 8.1 Hz, 2H), 6.70 (dd, J = 7.7, 4.9 Hz, 1H), 6.26 (s, 1H), 5.94 (s, 2H), 4.03 (s, 2H), 3.97 (s, 2H). MS: 343.2 $[\text{M}+\text{H}]^+$.

Example 10 (for reference only): Synthesis of 3-(3-(4-((1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

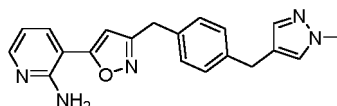
[0140]



[0141] The title compound was prepared according to the procedure described in Example 8 using (1H-pyrazol-4-yl)boronic acid (56.0mg, 0.50mmol) and heating for 14h at 90°C to yield 3-(3-(4-((1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (40mg, 0.12mmol, 36%) as a white solid. MS: 332.3 $[\text{M}+\text{H}]^+$.

Example 11 (for reference only): Synthesis of 3-(3-(4-((1-methyl-1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

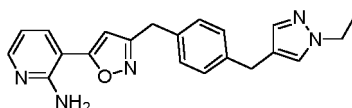
[0142]



[0143] The title compound was prepared according to the procedure described in Example 8 using (1-methyl-1H-pyrazol-4-yl)boronic acid (63.0mg, 0.50mmol) to yield 3-(3-(4-((1-methyl-1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (60mg, 0.17mmol, 52%) as a white solid. MS: 346.2 $[\text{M}+\text{H}]^+$.

Example 12 (for reference only): Synthesis of 3-(3-(4-((1-ethyl-1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

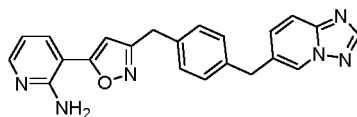
[0144]



[0145] The title compound was prepared according to the procedure described in Example 8 using (1-ethyl-1H-pyrazol-4-yl)boronic acid (70.0mg, 0.50mmol) to yield 3-(3-(4-((1-ethyl-1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (76mg, 0.21mmol, 63%) as a white solid. MS: 360.3 $[\text{M}+\text{H}]^+$.

Example 13 (for reference only): Synthesis of 3-(3-(4-([1,2,4]triazolo[1,5-a]pyridin-6-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine

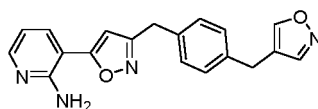
[0146]



[0147] The title compound was prepared according to the procedure described in Example 8 using 6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-[1,2,4]triazolo[1,5-a]pyridine (180mg, 0.73mmol) to yield 3-(3-(4-([1,2,4]triazolo[1,5-a]pyridin-6-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (85mg, 0.22mmol, 61%) as a white solid. MS: 383.3 [M+H]⁺.

Example 14 (for reference only): Synthesis of 3-(3-(4-(isoxazol-4-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine

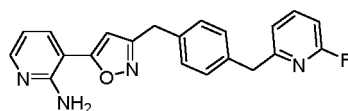
[0148]



[0149] The title compound was prepared according to the procedure described in Example 8 using isoxazol-4-ylboronic acid (56.5mg, 0.50mmol) to yield 3-(3-(4-(isoxazol-4-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (76mg, 0.23mmol, 69%) as a white solid. MS: 333.2 [M+H]⁺.

Example 15 (for reference only): Synthesis of 3-(3-(4-((6-fluoropyridin-2-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

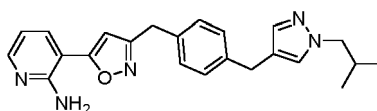
[0150]



[0151] The title compound was prepared according to the procedure described in Example 8 using (6-fluoropyridin-2-yl)boronic acid (70.5mg, 0.50mmol) to yield 3-(3-(4-((6-fluoropyridin-2-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (82mg, 0.23mmol, 68%) as a white solid. 500 MHz ¹H NMR (DMSO-d₆) δ 8.08 (dd, J = 4.8, 1.8 Hz, 1H), 7.93 - 7.83 (m, 2H), 7.29 - 7.20 (m, 5H), 6.97 (dd, J = 8.1, 2.6 Hz, 1H), 6.80 (s, 1H), 6.69 (dd, J = 7.7, 4.8 Hz, 1H), 6.24 (s, 2H), 4.01 (s, 2H), 3.99 (s, 2H). MS: 361.4 [M+H]⁺.

Example 16 (for reference only): Synthesis of 3-(3-(4-((1-isobutyl-1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

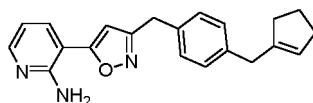
[0152]



[0153] The title compound was prepared according to the procedure for described in Example 8 using (1-isobutyl-1H-pyrazol-4-yl)boronic acid (95.0mg, 0.57mmol) to yield 3-(3-(4-((1-isobutyl-1H-pyrazol-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (72mg, 0.19mmol, 56%) as a white solid. MS: 388.3 [M+H]⁺.

Example 17 (for reference only): Synthesis of 3-(3-(4-(cyclopent-1-en-1-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine

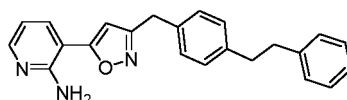
[0154]



[0155] The title compound was prepared according to the procedure described in Example 8 using cyclopent-1-en-1-ylboronic acid (63.5mg, 0.57mmol) to yield 3-(3-(4-(cyclopent-1-en-1-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine (72mg, 0.22mmol, 65%) as a white solid. MS: 332.2 [M+H]⁺.

Example 18 (for reference only): Synthesis of 3-(3-(4-phenethylbenzyl)isoxazol-5-yl)pyridin-2-amine

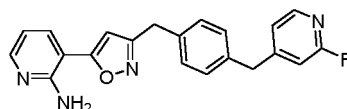
[0156]



[0157] The title compound was prepared according to the procedure described in Example 8 using 2-benzyl-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (124mg, 0.57mmol) to yield 3-(3-(4-phenethylbenzyl)isoxazol-5-yl)pyridin-2-amine (41mg, 0.12mmol, 35%) as a white solid. MS: 356.2 [M+H]⁺.

Example 19 (for reference only): Synthesis of 3-(3-(4-((2-fluoropyridin-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

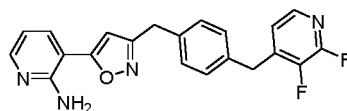
[0158]



[0159] The title compound was prepared according to the procedure described in Example 8 using (2-fluoropyridin-4-yl)boronic acid (70.5mg, 0.50mmol) to yield 3-(3-(4-((2-fluoropyridin-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (86mg, 0.24mmol, 72%) as a white solid. MS: 346.2 [M+H]⁺.

Example 20 (for reference only): Synthesis of 3-(3-(4-((2,3-difluoropyridin-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine

[0160]



[0161] The title compound was prepared according to the procedure described in Example 8 using (2,3-difluoropyridin-4-yl)boronic acid (106mg, 0.67mmol) to yield 3-(3-(4-((2,3-difluoropyridin-4-yl)methyl)benzyl)isoxazol-5-yl)pyridin-2-amine (85mg, 0.23mmol, 67%) as a white solid. MS: 379.2 [M+H]⁺.

Example 21 (for reference only): Synthesis of 3-(3-(4-(pyrimidin-5-ylmethyl)benzyl)isoxazol-5-yl)pyridin-2-amine

[0162]